Table 1: Positioning of sGSVA relative to representative single-sample gene-set scorers and pathway-aware spatial methods. Filled circles denote a property that is fully present, half circles a property present with caveats, open circles a property that is absent, and dashes a property that does not apply because the method addresses a different task. sGSVA is the only method that satisfies all six properties jointly; its distinctive contribution is the conjunction rather than any single column.

Method	Reference-free (no matched scRNA-seq)	Models spatial neighbourhood	Signed / directionality-aware (data-driven up/down)	Output is a per-location pathway score matrix	Built-in spatial significance testing	Spot- *and* cell-resolution
Single-sample gene-set scorers						
GSVA[a]	●	○	○	●	○	◐
ssGSEA	●	○	○	●	○	◐
AUCell	●	○	○	●	○	◐
UCell	●	○	○	●	○	◐
AddModuleScore	●	○	○	●	○	◐
Pathway-aware spatial methods						
GSDensity	●	◐	○	◐	◐	◐
Path-MGCN[b]	●	●	○	○	○	◐
EnrichMap[c]	●	●	○	●	◐	◐
PASTA[d]	○	●	○	◐	○	◐
Topology-aware (PSF)[e]	●	●	◐	◐	○	◐
scGIST[f]	–	–	–	–	–	–
sGSVA	●	●	●	●	●	●

● present ◐ partial / with caveats ○ absent – not applicable (different task)

[a] GSVA computes a bidirectional enrichment statistic but does not learn data-driven up/down gene subsets per pathway; "signed" here denotes data-driven directional contrast. The remaining single-sample scorers are one-sided.

[b] Path-MGCN (and related models such as PathCLAST) output spatial domain labels; pathway activity is an internal representation rather than the deliverable.

[c] EnrichMap is the nearest method on the spatial-scoring axes (reference-free, spatial, per-location score) but does not perform data-driven signed scoring.

[d] PASTA requires a matched scRNA-seq reference for pathway-expression imputation.

[e] Topology-aware / Pathway Signal Flow encodes signalling directionality through pathway topology rather than data-driven up/down subsets learned from expression.

[f] scGIST addresses gene-panel design, a complementary upstream task; included for completeness.